DML — Data Manipulation Language

Introduction to DML

What is DML?

DML stands for **Data Manipulation Language**. It is a subset of SQL used to manage, manipulate, and modify data and rows inside an existing database table.

Unlike DDL (Data Definition Language) which deals with the structure of the database (like creating or altering tables), DML focuses purely on the data stored within those tables.

Main DML Commands

We will cover the three core DML commands:

1. **INSERT** — Used to add new rows of data.
2. **UPDATE** — Used to modify existing rows of data.
3. **DELETE** — Used to remove existing rows of data.

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1. INSERT

Definition of INSERT

The `INSERT` command is used to add new rows (records) into a table.

Basic Syntax

```
INSERT INTO table_name (column1, column2, column3, ...)
VALUES (value1, value2, value3, ...);
```

Multiple Examples

Example 1: Insert a complete row

```
INSERT INTO student
VALUES (1, 'Rahul', 22, 'Delhi', 75);
```

Explanation: This inserts a full record for a student matching all five columns of the table in their default order.

Example 2: Specify column names

```
INSERT INTO student (id, name, age, city, marks)
VALUES (2, 'Amit', 23, 'Mumbai', 82);
```

Explanation: Here, the column names are explicitly specified, making the query clear and independent of the table's default column sequence as long as the values match the specified columns.

Example 3: Insert only selected columns

```
INSERT INTO student (id, name, marks)
VALUES (3, 'Priya', 21);
```

Explanation: This inserts data only into the `id`, `name`, and `marks` columns.

> **Note on unspecified columns:** Columns that are not specified (such as `age` and `city` in this example) will automatically receive `NULL` or their defined `DEFAULT` value, provided they allow it.

Example 4: Insert multiple rows

```
INSERT INTO student
VALUES
(4, 'Ravi', 24, 'Pune', 68),
(5, 'Sneha', 20, 'Delhi', 91),
(6, 'Karan', 22, 'Kolkata', 78);
```

Explanation: MySQL allows inserting multiple rows in a single statement by separating each set of values with commas, which improves performance and reduces code.

Example 5: Multiple rows with selected columns

```
INSERT INTO student (id, name, age, city, marks)
VALUES
(7, 'Neha', 21, 'Hyderabad', 88),
(8, 'Vikas', 23, 'Chennai', 72);
```

Explanation: Inserts multiple rows while explicitly targeting specific column names for all rows in the batch.

Important Points on INSERT

- `INSERT` creates new rows.
- `INSERT` does not modify an existing row.
- If column names are not specified, values follow the table's column order.
- If only selected columns are supplied, unspecified columns may receive `NULL` or their `DEFAULT` value, when allowed.
- MySQL supports inserting multiple rows using one `INSERT` statement.

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2. UPDATE

Definition of UPDATE

The `UPDATE` command is used to modify or change existing data values within the rows of a table.

Basic Syntax

```
UPDATE table_name
SET column1 = value1, column2 = value2, ...
WHERE condition;
```

Multiple Examples

Example 1: Update one column

```
UPDATE student
SET marks = 95
WHERE id = 1;
```

Explanation: Changes the `marks` to 95 specifically for the student whose `id` is 1.

Example 2: Update multiple columns

```
UPDATE student
SET age = 23,
    city = 'Hyderabad'
WHERE id = 2;
```

Explanation: Updates both the `age` and `city` columns simultaneously for the student with `id` = 2.

Example 3: Update multiple matching rows

```
UPDATE student
SET city = 'Delhi'
WHERE city = 'New Delhi';
```

Explanation: Updates the city to 'Delhi' for all rows where the current city is 'New Delhi'.

Example 4: Update using an existing value

```
UPDATE student
SET marks = marks + 5
WHERE marks < 50;
```

Explanation: Increases the existing `marks` value by 5 for every student who currently has marks below 50.

Example 5: UPDATE without WHERE

```
UPDATE student
SET city = 'Hyderabad';
```

Explanation: Updates the city to 'Hyderabad' for **every single row** in the table because no `WHERE` clause was specified.

Important Points on UPDATE

- `WHERE` is not compulsory in `UPDATE` syntax.
- `WHERE` is used to identify which rows should be updated.
- If `WHERE` is omitted, every row is eligible for the `UPDATE`.
- **Note:** If a classroom/database client blocks an update without a key-based `WHERE`, explain that this is a safety setting/tool behavior and not a general SQL rule.

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3. DELETE

Definition of DELETE

The `DELETE` command is used to remove existing rows from a table. The table structure, columns, and constraints remain intact.

Basic Syntax

```
DELETE FROM student
WHERE condition;
```

Multiple Examples

Example 1: Delete one row

```
DELETE FROM student
WHERE id = 3;
```

Explanation: Removes the student record where the `id` is 3.

Example 2: Delete rows matching a condition

```
DELETE FROM student
WHERE city = 'Delhi';
```

Explanation: Deletes all student records where the city is 'Delhi'.

Example 3: Delete using comparison

```
DELETE FROM student
WHERE marks < 40;
```

Explanation: Removes all students who scored less than 40 marks.

Example 4: Delete selected IDs

```
DELETE FROM student
WHERE id IN (2, 4, 7);
```

Explanation: Deletes multiple specific rows whose IDs match the values inside the `IN` list.

Example 5: DELETE without WHERE

```
DELETE FROM student;
```

Explanation: This removes all rows from the table, but the table itself remains (empty).

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4. UPDATE vs ALTER

Explanation

- **ALTER** is a **DDL** (Data Definition Language) command. It changes the table structure (such as adding, dropping, or modifying columns).
- **UPDATE** is a **DML** (Data Manipulation Language) command. It changes existing data/values inside the rows.

Examples

ALTER Example:

```
ALTER TABLE student
ADD email VARCHAR(100);
```

UPDATE Example:

```
UPDATE student
SET marks = 90
WHERE id = 1;
```

Comparison Table

Category	Purpose	What it changes	Example	Changes Structure?	Changes Row Data?
ALTER	Structural modification	Table schema/ columns	`ALTER TABLE student ADD email VARCHAR(100);`	Yes	No
UPDATE	Data modification	Row values	`UPDATE student SET marks = 90 WHERE id = 1;`	No	Yes

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5. DELETE vs TRUNCATE vs DROP

Explanation of Commands

- **DELETE:** DML | `DELETE FROM student WHERE id = 2;` | Removes rows, can use `WHERE`, table structure remains.
- **TRUNCATE:** DDL (in MySQL) | `TRUNCATE TABLE student;` | Removes all rows instantly, `WHERE` cannot be used, table structure remains.
- **DROP:** DDL | `DROP TABLE student;` | Removes the entire table itself along with data and structure, `WHERE` cannot be used.

Comparison Table

Command	Category	What it removes	Structure Remains?	WHERE allowed?	Selected rows?	Main purpose
DELETE	DML	Rows (all/ filtered)	Yes	Yes	Yes	Remove specific or all rows selectively
TRUNCATE	DDL (MySQL)	All rows	Yes	No	No	Quickly clear all data from a table
DROP	DDL	Entire table	No	No	No	Delete the table completely

> **Important MySQL Point:** `TRUNCATE` is classified as DDL in MySQL.

Questions to Ask While Teaching

DML basics

- What is DML?
- Why do we use DML?
- What are the main DML commands?
- Does DML work with table structure or table data?
- What is the difference between INSERT, UPDATE and DELETE?

INSERT

- How do you insert one complete row?
- Why can we specify column names in INSERT?
- What happens to columns omitted from INSERT?
- Can MySQL insert multiple rows using one INSERT?
- What happens if an inserted value violates a PRIMARY KEY or another constraint?

UPDATE

- What is UPDATE used for?
- Can UPDATE change more than one column?
- Can UPDATE affect more than one row?
- Is WHERE compulsory in UPDATE?
- What happens if WHERE is omitted?
- What is the difference between UPDATE and ALTER?

DELETE

- What is DELETE used for?
- Can DELETE remove selected rows?
- Can DELETE remove multiple rows?
- What happens with `DELETE FROM student;`?
- Does DELETE remove the table structure?

DELETE vs TRUNCATE vs DROP

- Which one is DML: DELETE, TRUNCATE or DROP?
- Which command can use WHERE?
- Which command removes all rows but keeps the table?
- Which command removes the table itself?
- Can WHERE be used with TRUNCATE?
- After `DROP TABLE student;`, can we insert into student immediately?
- After `TRUNCATE TABLE student;`, can we insert into student?
- What is the main purpose of DELETE compared with TRUNCATE?

Scenario questions

- You want to change only Rahul's marks. Which command do you use?
- You want to add a new student. Which command?
- You want to remove one student using ID. Which command?
- You want to remove all rows but keep the table. Which command?
- You want to remove the table itself. Which command?
- You want to add a new column to the table. Which command?

Quick Classroom Exercise

Give students these tasks:

1. Create/use a student table with `id`, `name`, `age`, `city` and `marks`.
2. Insert at least 5 students.

3. Update the marks of one student.
4. Update two columns of another student.
5. Update all students from one city to another city.
6. Delete one student using ID.
7. Delete students satisfying a condition.
8. Explain what happens if DELETE is used without WHERE.
9. Explain the difference between ALTER and UPDATE.
10. Explain DELETE vs TRUNCATE vs DROP.

Final Blackboard Summary

```
DML = Data Manipulation Language
INSERT → Add new rows
UPDATE → Modify existing rows
DELETE → Remove existing rows

ALTER vs UPDATE:
  ALTER changes structure; UPDATE changes row data.

DELETE vs TRUNCATE vs DROP:
  DELETE removes rows;
  TRUNCATE removes all rows but keeps the table;
  DROP removes the table itself.
```